CS341 · COMPUTER GRAPHICS

Calculations Solutions

Practice on every calculation type

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01 Solutions Q1–Q3

Q1

a) $AR = \dfrac{\frac{768}{9}}{\frac{1024}{12}} = \dfrac{85.33}{85.33} = 1$

b) $AR = \dfrac{\frac{600}{8}}{\frac{800}{10}} = \dfrac{75}{80} = 0.9375$

Q2

x: $\dfrac{1280}{32} = 40$ pixels per cm

y: $\dfrac{960}{24} = 40$ pixels per cm

$AR = \dfrac{40}{40} = 1$

Q3

pixels = 1024 × 768 = 786,432

a) 786,432 bits ÷ 8 = 98,304 bytes

b) 786,432 × 8 bits ÷ 8 = 786,432 bytes (768 KB)

c) 786,432 × 24 bits ÷ 8 = 2,359,296 bytes (2.25 MB)

02 Solution Q4: vectors

a) $v = Q - P = (7 - 2,\ 1 - 5) = (5, -4)$ $|v| = \sqrt{(25 + 16)} = \sqrt{41} \approx 6.4$

b) $|a| = \sqrt{(1 + 4 + 4)} = 3$ $\hat{a} = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

c) $a \cdot b = 1(2) + 2(-1) + 2(0) = 0$ → perpendicular

d) $a \times b = (a_y \cdot b_z - a_z \cdot b_y,\ a_z \cdot b_x - a_x \cdot b_z,\ a_x \cdot b_y - a_y \cdot b_x)$

$= (2 \cdot 0 - 2 \cdot (-1),\ 2 \cdot 2 - 1 \cdot 0,\ 1 \cdot (-1) - 2 \cdot 2)$

$= (2, 4, -5)$

check: $(2, 4, -5) \cdot a = 2 + 8 - 10 = 0$ ✓

e) $u \cdot w = -1$ $|u| = 1$ $|w| = \sqrt{2}$

$\cos\theta = -\frac{1}{\sqrt{2}}$ → $\theta = 135°$

f) $c_\perp = (-c_y, c_x) = (2, 3)$

03 Solution Q5: points and lines

Combination	Sum of coefficients	Valid point?
0.5P + 0.5R	1	**Yes (midpoint)**
2P − R	1	**Yes**
P + R	2	**No**
0.2P + 0.3R	0.5	**No**

```
b) P = (1, 2) + 0.75(8, 4) = (1 + 6, 2 + 3) = (7, 5)
```

```
c) b = B − C = (3, 4)          L(t) = (2, 1) + t(3, 4)
```

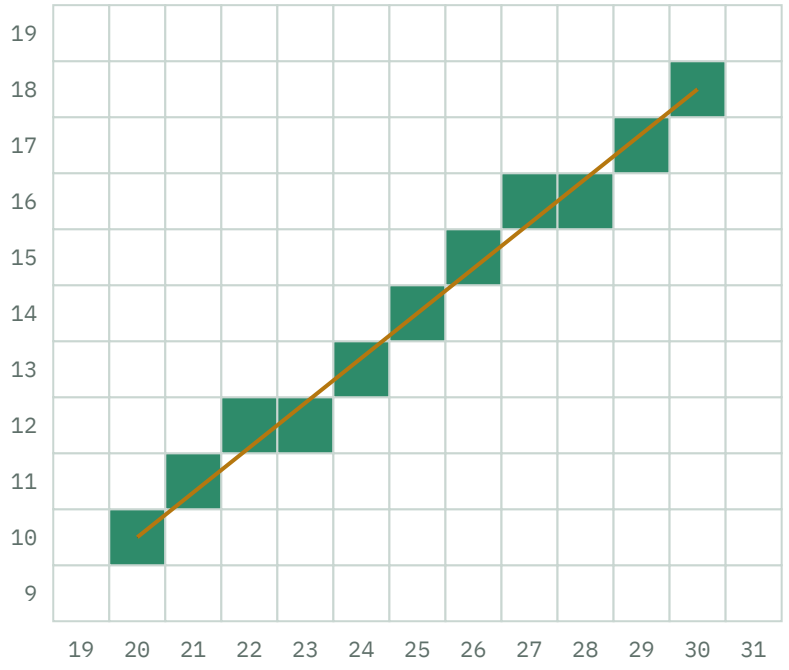
```
   n = b⊥ = (−4, 3)            −4(x − 2) + 3(y − 1) = 0
```

04 Solutions Q6–Q8: DDA

Q6: (20,10) → (30,18)

```
dx = 10, dy = 8, steps = 10, xInc = 1, yInc = 0.8
```

k	x	y	round(x)	round(y)
0	20	10	20	10
1	21	10.8	21	11
2	22	11.6	22	12
3	23	12.4	23	12
4	24	13.2	24	13
5	25	14	25	14
6	26	14.8	26	15
7	27	15.6	27	16
8	28	16.4	28	16
9	29	17.2	29	17
10	30	18	30	18

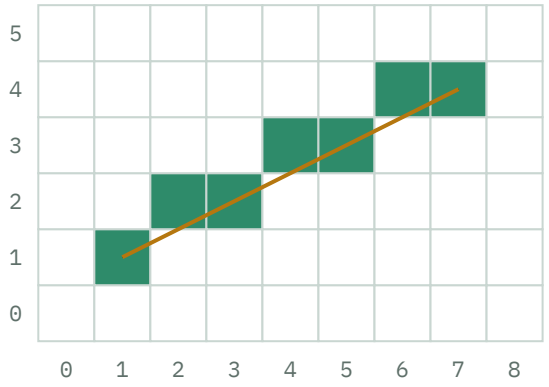


Q7: (1,1) → (7,4)

$$dx = 6,\ dy = 3,\ steps = 6,\ xInc = 1,\ yInc = \frac{3}{6} = 0.5$$

$$round(a) = int(a + 0.5) \quad \rightarrow \quad 1.5 \rightarrow 2, \quad 2.5 \rightarrow 3, \quad 3.5 \rightarrow 4$$

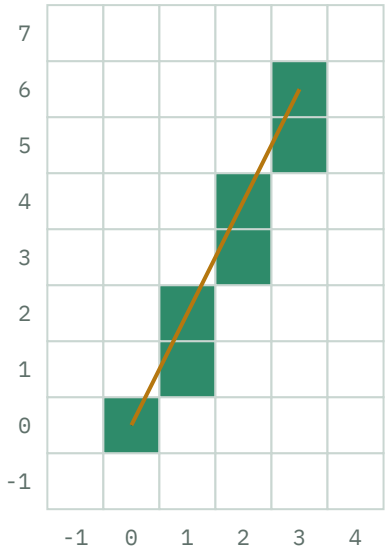
k	x	y	round(x)	round(y)
0	1	1	1	1
1	2	1.5	2	2
2	3	2	3	2
3	4	2.5	4	3
4	5	3	5	3
5	6	3.5	6	4
6	7	4	7	4



Q8: (0,0) → (3,6)

$$dx = 3,\ dy = 6,\ m = 2 > 1,\ steps = 6,\ xInc = \frac{3}{6} = 0.5,\ yInc = 1$$

k	x	y	round(x)	round(y)
0	0	0	0	0
1	0.5	1	1	1
2	1	2	1	2
3	1.5	3	2	3
4	2	4	2	4
5	2.5	5	3	5
6	3	6	3	6

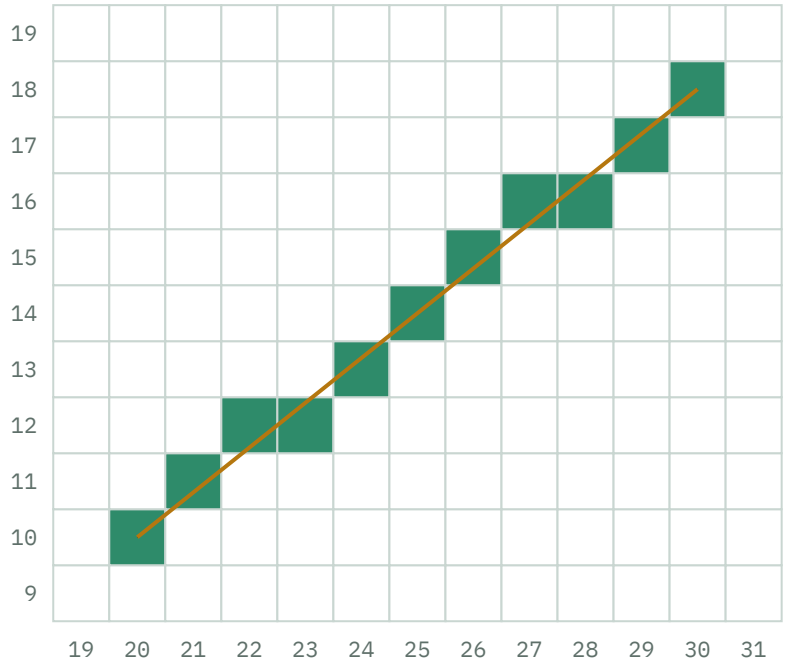


05 Solutions Q9–Q12: Bresenham

Q9: (20,10) → (30,18)

$\Delta x = 10,\ \Delta y = 8,\ 2\Delta y = 16,\ 2\Delta y - 2\Delta x = -4,\ p0 = 6$

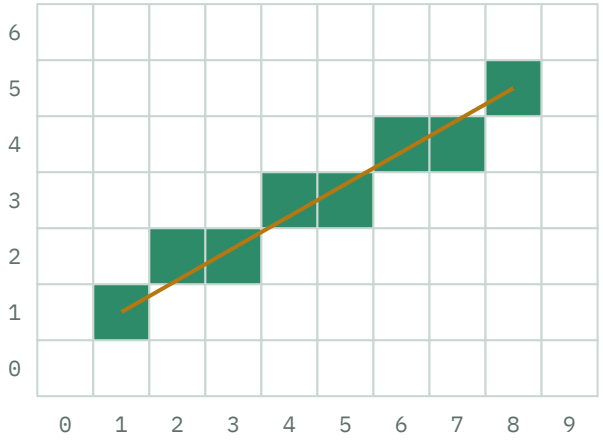
k	p_k	Decision	Next pixel	p_{k+1}
0	6	$p_k \geq 0$	(21, 11)	2
1	2	$p_k \geq 0$	(22, 12)	-2
2	-2	$p_k < 0$	(23, 12)	14
3	14	$p_k \geq 0$	(24, 13)	10
4	10	$p_k \geq 0$	(25, 14)	6
5	6	$p_k \geq 0$	(26, 15)	2
6	2	$p_k \geq 0$	(27, 16)	-2
7	-2	$p_k < 0$	(28, 16)	14
8	14	$p_k \geq 0$	(29, 17)	10
9	10	$p_k \geq 0$	(30, 18)	6



Q10: (1,1) → (8,5)

$\Delta x = 7,\ \Delta y = 4,\ 2\Delta y = 8,\ 2\Delta y - 2\Delta x = -6,\ p0 = 1$

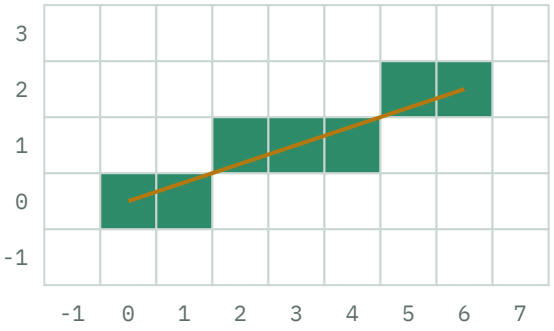
k	p_k	Decision	Next pixel	p_{k+1}
0	1	$p_k \geq 0$	(2, 2)	-5
1	-5	$p_k < 0$	(3, 2)	3
2	3	$p_k \geq 0$	(4, 3)	-3
3	-3	$p_k < 0$	(5, 3)	5
4	5	$p_k \geq 0$	(6, 4)	-1
5	-1	$p_k < 0$	(7, 4)	7
6	7	$p_k \geq 0$	(8, 5)	1



Q11: (0,0) → (6,2)

$\Delta x = 6,\ \Delta y = 2,\ 2\Delta y = 4,\ 2\Delta y - 2\Delta x = -8,\ p0 = 4 - 6 = -2$

k	p_k	Decision	Next pixel	p_{k+1}
0	-2	$p_k < 0$	(1, 0)	2
1	2	$p_k \geq 0$	(2, 1)	-6
2	-6	$p_k < 0$	(3, 1)	-2
3	-2	$p_k < 0$	(4, 1)	2
4	2	$p_k \geq 0$	(5, 2)	-6
5	-6	$p_k < 0$	(6, 2)	-2



Q12: (2,3) → (5,10) m > 1

The slope $\frac{7}{3} > 1$, so x and y swap roles: step in y; `p < 0` → `(x, y+1)`, `p ≥ 0` → `(x+1, y+1)`.

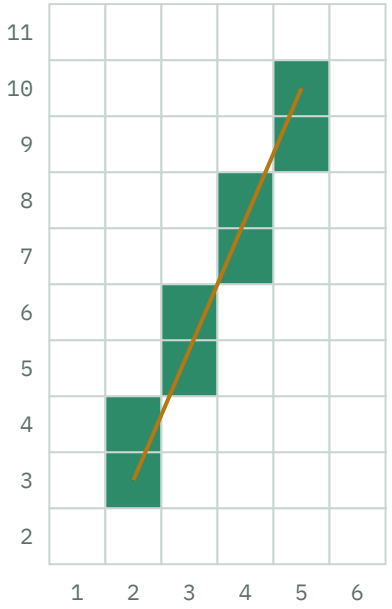
```
Δx = 3, Δy = 7
```

```
p0 = 2Δx − Δy = 6 − 7 = −1
```

```
p < 0:  p = p + 2Δx = p + 6
```

```
p ≥ 0:  p = p + 2Δx − 2Δy = p − 8
```

k	p_k	Decision	Next pixel	p_{k+1}
0	-1	$p_k < 0$	(2, 4)	5
1	5	$p_k \geq 0$	(3, 5)	-3
2	-3	$p_k < 0$	(3, 6)	3
3	3	$p_k \geq 0$	(4, 7)	-5
4	-5	$p_k < 0$	(4, 8)	1
5	1	$p_k \geq 0$	(5, 9)	-7
6	-7	$p_k < 0$	(5, 10)	-1



06 Solutions Q13–Q14: circles

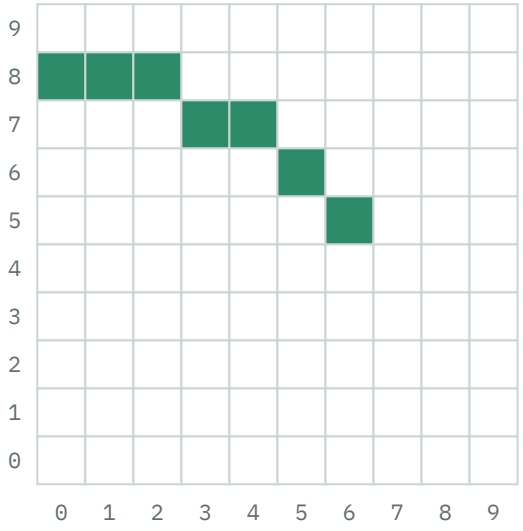
Q13: r = 8

```
start (0, 8),  d0 = 3 − 2(8) = −13
```

```
d < 0:  (x+1, y)      d = d + 4x + 6
```

```
d ≥ 0:  (x+1, y−1)    d = d + 4(x − y) + 10
```

k	d_k	Case	Pixel	d_{k+1}
start	—	$d_0 = 3 - 2r$	(0, 8)	-13
0	-13	d < 0 → T	(1, 8)	-7
1	-7	d < 0 → T	(2, 8)	3
2	3	d ≥ 0 → S	(3, 7)	-11
3	-11	d < 0 → T	(4, 7)	7
4	7	d ≥ 0 → S	(5, 6)	5
5	5	d ≥ 0 → S	(6, 5)	11



Q14: r = 10

start (0, 10), d0 = 3 − 2(10) = −17

k	d_k	Case	Pixel	d_{k+1}
start	—	$d_0 = 3 - 2r$	(0, 10)	-17
0	-17	d < 0 → T	(1, 10)	-11
1	-11	d < 0 → T	(2, 10)	-1
2	-1	d < 0 → T	(3, 10)	13
3	13	d ≥ 0 → S	(4, 9)	-5
4	-5	d < 0 → T	(5, 9)	17
5	17	d ≥ 0 → S	(6, 8)	11
6	11	d ≥ 0 → S	(7, 7)	13

Octant: (0,10) (1,10) (2,10) (3,10) (4,9) (5,9) (6,8) (7,7)

Symmetric points of (3, 10):

(3, 10) (10, 3) (−3, 10) (−10, 3)

(3, −10) (10, −3) (−3, −10) (−10, −3)